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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

```
1 import numpy as np
2 row, cols = map(int, input().split())
3 elements = []
4 for _ in range(row):
5     elements.extend(map(int, input().split()))
6 arr = np.array(elements).reshape(row, cols)
7
8 print(arr)
9 print(arr.ndim)
10 print(arr.shape)
11 print(arr.size)
12
```

Terminal Test cases

Debugger

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3.2.1. Numpy: Matrix Operations

23/53

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition (`matrix_a + matrix_b`)
2. Subtraction (`matrix_a - matrix_b`)
3. Element-wise Multiplication (`matrix_a * matrix_b`)
4. Matrix Multiplication (`matrix_a . matrix_b`)
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

matrixOp...

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1 `import numpy as np`

2

3 `# Input matrices`

4 `print("Enter Matrix A:")`

5 `matrix_a = np.array([list(map(int, input().split())) for i in range(3)])`

6

7 `print("Enter Matrix B:")`

8 `matrix_b = np.array([list(map(int, input().split())) for i in range(3)])`

9

10

11 `# Addition`

12 `print("Addition (A + B):")`

13 `print(matrix_a + matrix_b)`

14 `# Subtraction`

15 `print("Subtraction (A - B):")`

16 `print(matrix_a - matrix_b)`

17 `# Multiplication (element-wise)`

18 `print("Element-wise Multiplication (A * B):")`

19 `print(matrix_a * matrix_b)`

20 `# Matrix multiplication (dot product)`

21 `print("A dot B:")`

22 `print(matrix_a . matrix_b)`

23 `# Transpose`

24 `print("Transpose of A:")`

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 result = np.hstack((arr1, arr2))
12 print("Horizontal Stack:")
13 print(result)
14 # Perform vertical stacking (vstack)
15 result = np.vstack((arr1, arr2))
16 print("Vertical Stack:")
17 print(result)
```

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3.2.3. Numpy: Custom Sequence Generation

03:04

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

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```
1 import numpy as np
2
3 # Take user input for the start, stop, and step of the sequence
4 start = int(input())
5 stop = int(input())
6 step = int(input())
7
8 # Generate the sequence using np.arange()
9 sequence = np.arange(start, stop, step)
10 # Print the generated sequence
11 print(sequence)
```

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitw... 06:14

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_1 OR B_1).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4
5     # Convert A and B to NumPy arrays
6     arr_A = np.array(A)
7     arr_B = np.array(B)
8     # Arithmetic Operations
9     sum_result = arr_A + arr_B
10    diff_result = arr_A - arr_B
11    prod_result = arr_A * arr_B
12
13    # Statistical Operations
14    mean_A = np.mean(arr_A)
15    median_A = np.median(arr_A)
16    std_dev_A = np.std(arr_A)
17
18    # Bitwise Operations
19    and_result = arr_A & arr_B
20    or_result = arr_A | arr_B
21    xor_result = arr_A ^ arr_B
22
23    # Output results with one space between each element
24    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
```

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3.2.5. Numpy: Copying and Viewing Arrays

04:59

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The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

Sample Test Cases

copyAnd...

SUBMIT

Debugger

```
1 import numpy as np
2
3 inputlist = list(map(int, input().split(" ")))
4
5 #Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
16 print("Original array after modifying view:", original_array)
17 print("View array:", view_array)
18
19 # Modify the copy
20 copy_array[1] = 88
21 print("Original array after modifying copy:", original_array)
22 print("Copy array:", copy_array)
23
```

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Test cases

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

08:35

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. **Searching:** Find the indices where search_value appears in array1 and print these indices.
2. **Counting:** Count how many times count_value appears in array1 and print the count.
3. **Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
4. **Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding broadcast_value to each element.
4. The sorted array1.

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Debugger

```
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split())))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
9 broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
12 a = np.where(array1 == search_value)[0]
13 print(a)
14 # Count occurrences in array1
15 b = np.count_nonzero(array1 == count_value)
16 print(b)
17 # Broadcasting addition
18 c = array1 + broadcast_value
19 print(c)
20 # Sort the first array
21 d = np.sort(array1)
22 print(d)
```

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3.2.7. Student Data Analysis and Operations

05:38

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.

Operation...

```
1 import numpy as np
2
3 a=np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
4
5 #1. Print all student details
6 print("All student Details:\n",a)
7
8 #2. print total students
9 print("Total Students:",a.shape[0])
10
11 #3. Print all student Roll numbers
12 print("All Student Roll Nos",a[:,0])
13
14 #4. Print subject 1 marks
15 print("Subject 1 Marks",a[:,1])
16
17 #5. print minimum marks of Subject 2
18 print("Min marks in Subject 2", np.min(a[:,2]))
19
20 #6. print maximum marks of Subject 3
21 print("Max marks in Subject 3", np.max(a[:,3]))
22
23 #7. Print All subject marks
24 print("All subject marks:", a[:,1:])
```

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